Whitepaper: Intelligent Multi-Provider Auto-Routing & Resilience Engine

---

1. Executive Summary

In distributed LLM applications, relying on a single provider or model creates vulnerability to rate limits (HTTP 429), regional outages (HTTP 503/500), and latency spikes. The **Auto Mode Routing Engine** is an adaptive, client-side load balancer and failure-mitigation system designed to:

- Dynamically route requests to high-speed models across configured providers.
- Integrate newly added models with zero cold-start starvation.
- Ensure equal load distribution among fast models without over-stressing new additions.
- Automatically isolate failing models using circuit breakers and re-benchmark degraded models via exploration cycles.

---

2. Core Architecture & Pipeline

When an LLM operation is triggered in **Auto Mode**, the router executes a five-stage selection pipeline:

codeCode

```
[ Incoming Request ]
         │
         ▼
[ 1. Discovery & Provider Validation ] ──► (Filter out unconfigured or
keyless providers)
         │
         ▼
[ 2. Circuit Breaker & Lockout Filter ] ──► (Exclude models currently in
cooldown lock)
         │
         ▼
[ 3. Metric Aggregation & Tier Classification ]
   ├── Tier 1 (Probe/New)   : 0 successes / untested
   ├── Tier 1 (Fast)        : < 15,000 ms response time
```

```
    ├── Tier 2 (Medium)      : 15,000 ms - 25,000 ms response time
    └── Tier 4 (Demoted)     : ≥ 25,000 ms response time
         |
         ▼
[ 4. Dynamic Candidate Routing Algorithm ]
    ├── Exploration Cycle (Every 12th request) ──▶ Sample Tier 2 or Tier 4
    ├── Probe Queue Priority                     ──▶ Rotate untested models
first (1 sample graduation)
    ├── Normal Tier 1 Rotation                  ──▶ Fair Round-Robin across
all Tier 1 Fast models
    └── Fallback Cascade                        ──▶ Tier 2 ──▶ Tier 4 ──▶
Failover
         |
         ▼
[ 5. Execution & Metric Telemetry Update ] ──▶ (Record response time, rolling
window, or lock on error)
```

---

3. Metric Tracking & Sliding Window

For every model candidate (`provider, model`), the engine tracks persistent telemetry in `localStorage`:

Telemetry State Schema

- **`totalCalls`**: Cumulative number of dispatched requests.
- **`totalSuccesses`**: Cumulative number of successful responses (HTTP 200).
- **`recentTimesMs`**: Rolling FIFO buffer of up to **10 successful response latencies**.
- **`lastResponseTimeMs`**: Arithmetic mean of `recentTimesMs` (excluding error durations).
- **`lastTestedAt`**: Epoch timestamp of the last attempt.
- **`lastError`**: Timestamp and error message of the most recent failure.

---

4. Performance Tier Definitions

Tier	Classification	Condition	Routing Behavior
Tier 1 (Probe/New)	High Priority (Probe)	`totalSuccesses == 0` or untested	**Probed first** in fair round-robin until 1

			successful response is recorded.
Tier 1 (Fast)	High Priority (Active)	`Effective Latency < 15s` & `totalSuccesses ≥ 1`	**Primary production pool**. Rotates with equal probability in round-robin order.
Tier 2 (Medium)	Medium Speed	`15s ≤ Effective Latency < 25s`	Standby pool; probed periodically every 12th request for promotion.
Tier 4 (Demoted)	Slower / Degrading	`Effective Latency ≥ 25s`	Emergency fallback; probed periodically during exploration cycles.

---

5. Candidate Selection Logic

5.1 Exploration Probing (
-Greedy Scheduling)

To prevent slower or recovering models from being permanently isolated, the engine maintains an exploration counter:

- If Tier 2 or Tier 4 candidates exist when the trigger fires, the engine routes the query to one of these models as an exploratory benchmark.
- If the exploratory request returns in

  , the rolling average shifts and the model automatically promotes back into **Tier 1 (Fast)**.

5.2 Probe/New Model Handling

- **Untested models** (0 successful calls recorded) enter the **Tier 1 Probe Queue**.
- Probing rotates across available untested models evenly using the global rotation index (

).

- **1-Sample Graduation**: As soon as a probe model records **1 successful response**, it graduates out of the probe queue and joins the standard **Tier 1 (Fast)** round-robin rotation.
- This eliminates the cold-start problem while preventing new models from being overwhelmed by cumulative call-count catching.

5.3 Normal Round-Robin Rotation

When no untested models are queued and the request is not an exploration cycle:

- The router cycles through all active **Tier 1 (Fast)** models evenly:

---

6. Circuit Breaker & Fault Recovery

When a model encounters a failure, the engine applies exponential backoff locks to prevent cascading failure loops:

codeCode

```
┌───────────────────────┐
│ Model Request Fails   │
└───────────┬───────────┘
            │
 ┌──────────┴──────────┐
 ▼                     ▼
[ Rate Limit (429) ]          [ Server Error (500/503) ]
Lock duration:                Lock duration:
1 Hour Cooldown               24 Hours Cooldown
 │                     │
 └──────────┬──────────┘
            ▼
┌────────────────────────┐
│ Model Marked as "Locked" │
│ Excluded from Selection  │
└───────────┬────────────┘
            │
 ┌──────────┴──────────┐
 ▼                     ▼
[ Cooldown Expires ]          [ Manual User Unlock ]
Lock cleared                  Lock cleared
```

```
(Retains previous metrics)        (Retains previous metrics)
            │                                  │
            └──────────────┬───────────────────┘
                           ▼
                ┌──────────────────┐
                │ Re-enters Active Tier │
                └──────────────────┘
```

- **Rate Limit (HTTP 429)**: Triggers a **1-hour lockout**.
- **Server/Quota Error (HTTP 500 / 503)**: Triggers a **24-hour lockout**.
- **Manual Override**: Users can immediately clear locks via the **Model Status Modal** (`Unlock Now`) or reset all metrics globally (`Reset All States`).

---

7. Mathematical Summary Table

Parameter	Value	Purpose
Fast Latency Threshold		Qualifies for Tier 1 standard rotation
Medium Latency Threshold		Qualifies for Tier 2 standby pool
Exploration Frequency	Every	Re-evaluates Tier 2 / Tier 4 models for promotion
Probe Graduation Criteria		Moves model from Probe/New to active Tier 1 rotation
Rolling Metric Window		Smooths out transient network spikes
Rate Limit Lockout		Protects against quota exhaustion loops
Server Error Lockout		